

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS

Friday 2nd May 2014

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY BE USED IN THIS EXAMINATION

Year 3 Courses

Convener: S. Viglas

External Examiners: A. Cohn, T. Field

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Consider a circuit-switched network with 9 links each with a bandwidth of 10 Mbps (Figure 1). How many users (say, N) can the network support simultaneously if half of the users want 50 Kbps and half of them want 450 Kbps? [3 marks]

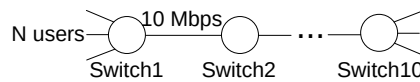


Figure 1: A simple circuit-switched network.

- (b) Assume that there is only one router between a source and destination. The bandwidth of the first link, i.e., from source to the router, is R_1 and that of the second link, i.e., from router to destination, is R_2 . What is the total end-to-end delay for a packet of length L . Ignore queuing, propagation, processing delays and store-and-forward. [2 marks]
- (c) Which of the two types of architectures—client-server or peer-to-peer—is more scalable in terms of number of nodes that can be supported in the architecture? Justify your answer. [2 marks]
- (d) Web caching traditionally has been designed to overcome the inherent propagation delays in the Internet by storing frequently retrieved content at the local cache. To help Web caching, the HTTP protocol uses the conditional GET which allows HTTP request messages to be tagged with “If-modified-since:” header line to fetch objects. Assume an institution’s access link has a capacity of 2.5 Gbps. Also, assume that GET requests, conditional GET requests and conditional GET responses (for the case when there is no modification of the object) are of size 1000 Bytes.
 - i. Suppose there are two object sizes that are most popular for the users in the institution. First is a small object of size 1 KByte and the second is a 50 MByte object. Assume that normal HTTP GET requests are less than 100 Bytes and both these types of objects are requested with equal probability (0.5). If the average request rate is 10 per second, what fraction of the access link’s bandwidth is utilised? [6 marks]
 - ii. Suppose we employ Web caching for the above scenario which has a cache hit ratio of 0.5 for objects. What will the access link utilisation be now? [6 marks]
 - iii. What if the cache hit rate for the small objects is 0.7 and that of the large objects is 0.1. Is it still beneficial to perform Web caching? [6 marks]

2. (a) Consider a p2p network which is used for file distribution. It has a hierarchy of peers, super peers and ultimate peers with each super peer responsible for 100 peers and each ultimate peer responsible for 100 super peers. The query propagation follows strict hierarchy, going all the way from the node to the super and then to the ultimate peer the super node is connected to. After that, the ultimate peer will contact other ultimate peers sequentially, until an answer is found. All ultimate peers are connected to each other.
- i. How many nodes exist in a network with 5 Million peers? *[3 marks]*
 - ii. Suppose that a query is answered by peers with a probability of 0.1, super peer with a probability of 0.2 and an ultimate peer with a probability of 0.5. What is the average number of messages required to answer a query? *[7 marks]*
- (b) Suppose there is a new application in the Internet which requires reliability but does not care about out-of-order delivery. What is the appropriate transport layer protocol for the application? TCP, UDP, or both? Justify your answer. *[2 marks]*
- (c) Describe one main difference between TCP Tahoe and TCP Reno. *[2 marks]*
- (d) Why is a checksum included in IPv4 datagrams when TCP and UDP segments already have checksum fields? *[2 marks]*
- (e) Explain the counting-to-infinity problem and describe a general mechanism for solving it. *[5 marks]*
- (f) Scalability of link state routing protocols, exemplified by OSPF, can be improved by making them hierarchical as with hierarchical OSPF. Suggest an alternate and more effective way to improve the scalability of link state protocols. *[4 marks]*

3. (a) Compare channel partitioning and random access MAC (medium access control) protocols at high and low loads. Which of these two classes of protocols permit statistical multiplexing and why? [6 marks]
- (b) Describe two scenarios where it is better to use *proactive* forward error correction instead of *reactive* error detection and retransmission. [4 marks]
- (c) An application has a message of size 65507 bytes to another directly connected host over a link with 1000 bytes maximum transmission unit (MTU). Assume that UDP and IPv4 are used as transport and network layer protocols, respectively. Also assume that acknowledgement packets in the stop-and-wait protocol are never lost or corrupted.
 - i. If the link has a 10 percent frame loss rate and uses the stop-and-wait protocol for loss recovery how many transmissions on average are needed to get the complete message across to the receiving host? [6 marks]
 - ii. Would there be any improvement if the application instead divided its message into smaller chunks so that each chunk can fit within a MTU? Explain your answer. [2 marks]
- (d) Twisted pair copper wiring used in Digital Subscriber Line (DSL) based broadband access networks is known to decay in signal quality with distance, and as a result, capacity of a DSL line also decreases with distance. Suppose that the capacity, c , of a DSL line (in Mbps) is modelled as a function, $c = 47.41e^{-0.6d}$, where d is the length of the local loop in kilometers (km).
 - i. How many central offices (telephone exchanges) are needed to serve broadband with a minimum download speed of 10Mbps to 25 houses uniformly distributed in a 1km x 1km area? [4 marks]
 - ii. How would your answer to the previous part change if the same 25 houses are spread uniformly over a 10km x 10km area instead? [3 marks]